

Course code: CSN 405
Course name: Network Programming
Type of Exam : Open book.

Exam code: 01
Duration : 90'
Maximum mark : 100

Instructions:

- *Non-programming calculator: Yes ☐ | No ☐*
- *This paper consists of 3 sections. This paper carries 100 marks*

Section 1 (30 marks) includes 6 questions. Each question carries 5 marks. No mark will be deducted for wrong answers.

Section 2 (30 marks) includes one question. A question carries 30 marks. No mark will be deducted for wrong answers.

Section 3 (40 marks) includes one question. A question carries 40 marks. No mark will be deducted for wrong answers.

Section 1/Question 1 (30 marks)

- Which method of the socket module allows a server socket to accept requests from a client socket from another host? (5 points)
- Which method of the socket module allows you to send data to a given address? (5 points)
- Which method of the socket module allows you to associate a host and a port with a specific socket? (5 points)
- What is the difference between the TCP and UDP protocols, and how do you implement them in Python with the socket module? (5 points)
- Which method of the socket module allows you to implement port scanning with sockets and to check the port state? (5 points)
- Describe why an application developer might choose to run an application over UDP rather than TCP. (5 points)

Section 2/ Question 2 (30 marks)

Design and implement a TCP server and client application in Python where clients attempt to guess a randomly generated number with feedback from the server in X times (number of guesses).

Requirements:

Server Implementation:

- Create a TCP server that listens on localhost, port 2026

- Generate a random number between 1 and 100 when a client connects
- For each client guess, provide one of the following responses:
 - “Too high! Try again.” if the guess is greater than the target
 - “Too low! Try again.” if the guess is less than the target
 - “Congratulations! You guessed correctly in X attempts!” when correct
- Track the number of attempts for each client
- Allow the client to disconnect gracefully

Client Implementation:

- Connect to the server on localhost, port 2026
- Prompt the user to enter their guess
- Send the guess to the server
- Display the server's response
- Continue until the correct number is guessed
- Display the total number of attempts upon winning

Deliverables

- *server.py* - The server implementation
- *client.py* - The client implementation
- Screenshots showing the inputs and outputs in the client and service sides.

Section 3/ Question 3 (40 marks)

Modify the server code in Section 2 so that it can interact with multiple clients using *multiple threads*. For each connection (using TCP sockets), the Server should provide the same service as described.

Deliverables

- *server.py* - The server implementation
- *client.py* - The client implementation
- Screenshots showing the inputs and outputs in the client and service sides.

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Lecturer's signature

Vũ Đức Lý